

BOONE COUNTY AP, AR, USA

WMO: 723450

Lat: 36.267N

Lon: 93.157W

Elev: 1374

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	16.1	0.1	5.8	16.0	4.5	7.3	20.8	23.4	34.1	20.7	40.8	7.9	310	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	95.1	74.2	92.1	74.3	89.9	73.8	77.6	88.5	76.5	87.2	75.4	85.9	7.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	136.0	83.1	73.3	130.1	81.7	72.4	126.1	80.8	42.0	88.7	40.8	87.1	39.8	85.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.8	16.1	4.7	98.2	3.7	4.4	2.0	101.4	-0.2	103.9	-2.3	106.4	-4.9	109.5		
			3.7	80.1	3.5	2.6	1.1	81.9	-0.9	83.5	-2.9	84.9	-5.5	86.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	37.4	40.9	49.1	58.9	66.5	74.7	78.6	77.9	70.3	59.8	49.1	39.8	(6)
(7)		DBStd	16.92	11.78	11.45	10.64	8.56	7.39	5.24	4.95	5.34	6.88	8.53	9.61	10.28	(7)
(8)		HDD50	1375	418	294	146	23	2	0	0	0	17	131	344		(8)
(9)		HDD65	3819	857	675	499	217	72	3	0	1	28	204	482	781	(9)
(10)		CDD50	4545	27	40	120	289	515	740	886	864	610	322	104	28	(10)
(11)		CDD65	1513	0	1	8	33	120	293	421	400	188	45	4	0	(11)
(12)		CDH74	14113	0	5	64	265	895	2663	4264	3939	1643	350	25	0	(12)
(13)		CDH80	5658	0	0	8	40	215	997	1930	1796	602	69	1	0	(13)
(14)	Wind	WSAvg	7.4	8.7	8.6	9.0	9.1	7.4	6.3	5.5	5.4	5.6	6.7	7.9	8.2	(14)
(15)	Precipitation	PrecAvg	45.6	2.4	2.8	4.0	4.3	5.2	4.9	3.3	3.5	4.1	3.5	4.1	3.4	(15)
(16)		PrecMax	73.8	5.6	7.4	11.9	9.8	12.9	9.8	9.5	9.9	11.3	10.4	12.4	8.9	(16)
(17)		PrecMin	26.7	0.3	0.8	0.6	0.6	1.4	1.1	0.9	0.0	0.4	0.2	0.4	0.3	(17)
(18)		PrecStd	10.0	1.6	1.7	2.5	2.4	2.4	2.2	1.8	2.0	2.7	2.8	2.8	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.9	73.6	80.3	84.1	89.1	94.7	99.4	99.5	94.7	85.5	77.5	69.6	(19)
(20)		MCWB	56.5	57.7	60.8	64.9	70.2	74.1	73.0	73.0	72.0	69.8	62.9	59.5		(20)
(21)		2%	DB	64.9	67.8	74.9	80.5	85.0	90.8	95.1	95.1	89.9	81.8	72.3	64.4	(21)
(22)		MCWB	55.5	56.1	59.7	63.2	70.0	73.8	74.5	74.1	71.6	66.9	59.2	56.4		(22)
(23)		5%	DB	60.9	63.7	70.5	77.0	82.3	88.3	91.8	92.3	86.7	78.3	68.7	60.7	(23)
(24)		MCWB	52.0	54.2	57.0	61.7	68.9	73.3	75.1	74.1	71.0	64.3	58.6	53.0		(24)
(25)		10%	DB	55.4	59.5	66.3	73.2	79.3	86.0	89.7	89.5	83.2	73.9	64.5	55.4	(25)
(26)		MCWB	48.0	50.8	55.3	60.2	67.3	72.6	74.7	73.6	69.1	62.9	55.8	48.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	62.3	64.4	68.6	74.4	78.0	79.2	79.1	75.7	72.5	65.9	62.4	(27)
(28)		MCDWB	66.3	69.0	75.8	79.0	84.0	88.9	90.1	89.7	86.7	81.1	72.9	67.0		(28)
(29)		2%	WB	58.0	59.7	61.9	66.5	72.4	76.4	77.9	77.3	74.2	70.1	62.9	58.9	(29)
(30)		MCDWB	63.0	65.4	71.5	75.6	81.6	87.1	88.7	88.3	84.6	77.5	68.9	63.5		(30)
(31)		5%	WB	53.6	56.0	59.5	64.8	70.8	75.0	76.9	76.1	72.9	67.6	60.2	54.5	(31)
(32)		MCDWB	58.2	61.9	67.5	73.1	79.3	85.1	87.5	87.2	82.6	74.1	66.3	58.8		(32)
(33)		10%	WB	48.4	51.2	57.0	62.9	69.1	73.7	75.7	74.9	71.4	65.3	57.3	50.1	(33)
(34)		MCDWB	54.5	58.1	65.0	70.6	76.9	83.1	86.0	85.6	80.1	71.9	62.9	54.2		(34)
(35)	Mean Daily Temperature Range	MDBR	18.3	18.9	20.5	20.9	19.1	19.3	19.5	20.2	21.3	20.9	19.4	17.4		(35)
(36)		5% DB	MCDBR	23.9	25.1	25.4	24.6	21.0	21.7	22.1	23.4	24.4	24.2	24.0	22.5	(36)
(37)		MCWBR	16.0	16.7	13.9	12.6	9.8	8.9	7.3	7.7	9.1	11.7	14.7	15.3		(37)
(38)		5% WB	MCDBR	20.7	22.7	21.5	20.3	18.6	19.3	19.2	20.0	20.3	20.1	20.2	20.3	(38)
(39)		MCWBR	16.4	17.4	14.2	12.4	9.6	9.2	8.0	7.9	8.8	11.4	15.0	16.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.304	0.338	0.378	0.409	0.425	0.453	0.451	0.385	0.342	0.313	0.292	(40)	
(41)		taud	2.506	2.497	2.434	2.324	2.290	2.265	2.244	2.221	2.390	2.494	2.527	2.545	(41)	
(42)		Ebn at Noon	286	298	295	287	278	272	264	262	276	280	279	282	(42)	
(43)		Edh at Noon	26	29	34	40	42	43	44	44	35	29	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	777	947	1285	1624	1787	2037	1977	1812	1551	1166	834	662	(44)	
(45)		RadStd	67	121	78	96	152	159	119	125	124	145	92	83	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page